

Vector Spaces

Math Foundations — Concept 05

1 Motivation

A vector space is the algebraic structure that captures “things you can scale and add together.” The motivating examples are arrows in $\mathbb{R}^n$, but the abstraction subsumes polynomial spaces $\mathbb{R}[x]$, function spaces $C([0, 1])$ and $L^2(\mathbb{R})$, sequence spaces ℓ^2 , and the embedding spaces $\mathbb{R}^d$ that store token representations inside transformers. Linear algebra is the study of these spaces and the structure-preserving maps between them.

2 Axioms

First, an example

Before stating the axioms abstractly, fix the prototype in your head. Take $V = \mathbb{R}^3$ with the field of scalars $\mathbb{F} = \mathbb{R}$. The *standard basis* is

$$\{e_1, e_2, e_3\} = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}.$$

Read aloud: the set containing one-zero-zero, zero-one-zero, and zero-zero-one. Every vector in $\mathbb{R}^3$ is a unique linear combination of these three. For instance, the vector $(2, 3, 5)$ decomposes as

$$(2, 3, 5) = 2e_1 + 3e_2 + 5e_3 = 2(1, 0, 0) + 3(0, 1, 0) + 5(0, 0, 1).$$

Read aloud: the vector two-three-five equals two times e-one plus three times e-two plus five times e-three. The coefficients $(2, 3, 5)$ are the *coordinates* of the vector with respect to the basis. The axioms below are simply the rules that make such decompositions consistent in any space.

Definition 1 (Vector space). Let $\mathbb{F}$ be a field (e.g. $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{F}_p$). A *vector space over $\mathbb{F}$* is a set V equipped with operations

$$+ : V \times V \rightarrow V, \quad \cdot : \mathbb{F} \times V \rightarrow V$$

Read aloud: plus, from V cross V to V ; and dot, from $\mathbb{F}$ cross V to V . such that $(V, +)$ is an abelian group and for all $\alpha, \beta \in \mathbb{F}$ and $u, v \in V$:

1. $\alpha(u + v) = \alpha u + \alpha v$,
2. $(\alpha + \beta)v = \alpha v + \beta v$,
3. $(\alpha\beta)v = \alpha(\beta v)$,
4. $1 \cdot v = v$.

Elements of V are called *vectors*; elements of $\mathbb{F}$ are *scalars*.

Definition 2 (Linear combination, span). Given $v_1, \dots, v_k \in V$ and $\alpha_1, \dots, \alpha_k \in \mathbb{F}$, the vector $\sum_i \alpha_i v_i$ is a *linear combination*. The *span* of a set $S \subseteq V$, denoted $\text{span}(S)$, is the set of all finite linear combinations of elements of S ; it is the smallest subspace containing S .

Definition 3 (Linear independence). A set $S \subseteq V$ is *linearly independent* if for every finite subset $\{v_1, \dots, v_k\} \subseteq S$, the equation $\sum_i \alpha_i v_i = 0$ implies $\alpha_1 = \dots = \alpha_k = 0$. Otherwise S is *linearly dependent*.

Definition 4 (Basis). A *basis* of V is a linearly independent set $B \subseteq V$ with $\text{span}(B) = V$. Equivalently, every $v \in V$ admits a *unique* expression as a finite linear combination of elements of B .

Definition 5 (Dimension). If V admits a finite basis, the *dimension* $\dim_{\mathbb{F}} V$ is the cardinality of any such basis (well-defined by Theorem 7 below). If no finite basis exists, V is *infinite-dimensional*.

3 The Dimension Theorem

Lemma 6 (Steinitz exchange). *Let V be a vector space, let $\{v_1, \dots, v_m\}$ be a spanning set, and let $\{w_1, \dots, w_n\}$ be linearly independent. Then $n \leq m$, and after relabeling we may replace n of the v_i by the w_j to obtain another spanning set $\{w_1, \dots, w_n, v_{n+1}, \dots, v_m\}$.*

Proof. Induct on n . The base case $n = 0$ is trivial. Suppose the result holds for $n - 1$: after relabeling, $\{w_1, \dots, w_{n-1}, v_n, \dots, v_m\}$ spans V . Since this set spans, write

$$w_n = \sum_{i=1}^{n-1} \alpha_i w_i + \sum_{j=n}^m \beta_j v_j.$$

Read aloud: w sub n equals the sum from i equals one to n minus one of α sub i times w sub i , plus the sum from j equals n to m of β sub j times v sub j . Not all β_j are zero, else $w_n \in \text{span}(w_1, \dots, w_{n-1})$, contradicting independence of $\{w_1, \dots, w_n\}$. In particular $m \geq n$. After relabeling, $\beta_n \neq 0$, so

$$v_n = \beta_n^{-1} \left(w_n - \sum_{i < n} \alpha_i w_i - \sum_{j > n} \beta_j v_j \right) \in \text{span}(w_1, \dots, w_n, v_{n+1}, \dots, v_m).$$

Read aloud: v sub n equals one over β sub n times the quantity w sub n minus the sum over i less than n of α sub i w sub i minus the sum over j greater than n of β sub j v sub j ; this lies in the span of w one through w n together with v n plus one through v m . Therefore $\{w_1, \dots, w_n, v_{n+1}, \dots, v_m\}$ spans V , completing the induction. $\square$

Theorem 7 (Dimension is well-defined). *Let V be a vector space and suppose V has a finite basis. Then any two bases of V have the same cardinality.*

Proof. Let $B_1 = \{v_1, \dots, v_m\}$ and $B_2 = \{w_1, \dots, w_n\}$ be two bases of V . Since B_1 spans V and B_2 is linearly independent, Lemma 6 gives $n \leq m$. By symmetry (swap the roles: B_2 spans, B_1 is independent), $m \leq n$. Therefore $m = n$. $\square$

Proposition 8 (Existence of a basis, finite case). *If V is spanned by finitely many vectors $v_1, \dots, v_m$, then some subset of $\{v_1, \dots, v_m\}$ is a basis of V .*

Proof sketch. If $\{v_1, \dots, v_m\}$ is linearly independent, done. Otherwise some v_k is a linear combination of the others; remove it. The remaining set still spans. Iterate; the process terminates because the set is finite, yielding a linearly independent spanning set. $\square$

4 Examples

Example 9 ($\mathbb{R}^3$). The standard basis $\{e_1, e_2, e_3\}$ with e_i the i -th coordinate vector gives $\dim \mathbb{R}^3 = 3$. The set $\{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}$ is another basis (the 3×3 matrix with these columns has determinant -2).

Example 10 (Polynomial space). $\mathbb{R}[x]_{\leq n} = \{a_0 + a_1x + \dots + a_nx^n : a_i \in \mathbb{R}\}$ has basis $\{1, x, x^2, \dots, x^n\}$, so $\dim = n + 1$.

Example 11 (Function space). $C([0, 1])$, the continuous real functions on $[0, 1]$, is infinite-dimensional: $\{1, x, x^2, x^3, \dots\}$ is linearly independent.

5 Connection to ML

In transformer-style models, every token is mapped to a vector in $\mathbb{R}^d$ ($d \approx 4096$ for large models). Attention computes weighted sums (linear combinations) of value vectors; layer norm, residual streams, and gradient updates all operate inside this vector space. The ELF paper (arxiv:2605.10938) defines flow-matching trajectories $x_t = (1 - t)x_0 + tx_1$ in $\mathbb{R}^d$, a literal application of the vector-space axioms.